D-Modem AT Command Reference

The modem core of slmodemd, libdmodem and the PBX modules

October 2026

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1 Introduction

D-Modem's modem core (`core/` in the source) takes the AT commands of every D-Modem program:

- **slmodemd**, the modem on a pseudo terminal (`/dev/ttySL0` links to it), its line a SIP call;
- **libdmodem** (`dmodem.h`), the modem as a library: the program that embeds it sends the commands;
- the **Asterisk and FreeSWITCH modules** (`app_dmodem`, `mod_dmodem`), whose profiles give the commands each call starts with.

The commands, and the way a command line is read, are those of Smart Link's slmodemd (Debian's `sl-modem-daemon`), whose ST56 modems' reference is kept as `docs/legacy/ST56ATCommands.pdf`. Section 10 lists where D-Modem differs from it, and section 9 what a SIP line changes in slmodemd.

This reference follows the core as it is: every command, parameter range, default and output here is the core's own, the outputs as slmodemd prints them. A setting that is kept but changes nothing says so.

Notation. `AT` commands are shown in capitals; the modem takes either case. n is a decimal number, h a hexadecimal one; `[ ]` marks what may be left out. A setting's default is the one `ATZ` and the start of the program give.

2 Command lines

- A command line begins with `AT` and ends with the character of S3 (carriage return) or S4 (line feed). Blanks and tabs in it are dropped before it is read.
- The character of S5 (backspace) takes back the last one typed.
- A line holds 63 characters at most; a longer one is ERROR.
- With echo on (`E1`), what the DTE types comes back as it is typed.
- A line that does not begin with `AT` is not a command: with echo on it is blanked out on the DTE's screen.
- `A/` (no line end needed) runs the last command line again.

The commands of a line run in turn. The first that fails ends the line with ERROR; otherwise OK follows the last one. `A`, `D` and `O` end the line and give their result later (CONNECT, NO CARRIER and so on), and `Z` ends it with OK: whatever follows them on the line is not read. A command that reports something (`I`, `S`*n*`?`, `+MS?` ...) prints it on lines of its own before the result.

A character from the DTE while a call is being set up (dialing, answering, the modems' training) hangs up.

3 Result codes

`V0`	`V1`	
0	OK	the command line ran
1	CONNECT [*rate*]	the call is up, or a fax command's data begins
2	RING	a call rings (slmodemd: every 3 s, section 9)
3	NO CARRIER	no call came up, or the call is over
4	ERROR	a command failed, or a line too long
6	NO DIALTONE	no dial tone (`X2`, `X4`)
7	BUSY	a busy tone
8	NO ANSWER	no answer within S7 seconds of dialing

`Q1` turns them off, `V0` gives the numbers, each on a line of its own. With `V1` a data call's CONNECT gives the rate the modem receives at, in bit/s (`CONNECT 33600`); with `X0`, in fax and in voice mode it is CONNECT alone.

Before CONNECT, `&A` (S70) adds lines on the connection (verbose results only), each bit one of them:

bit	line	
0 (1)	`Modulation:` *name*	the modulation's name (`V92`, `V34`, `V32bis`, ...)
1 (2)	`Protocol:` `LAPM` or `NONE`	V.42 error correction
0 and 1 (3)	`Compression:` `V44`, `V42bis` or `NONE`	what V.42 agreed
2 (4)	`TxRate:` *rate*	the rate the modem sends at

4 Basic commands

`ATA`

Answers: off hook, and the answering modem's tones after S92 seconds. In slmodemd the call that rings (`RING`) is answered. Its result is CONNECT, or NO CARRIER.

`ATB`

V.22 at 1200 bit/s only, without automode: `AT+MS=22,0,1200,1200`. (Not V.250's choice between the ITU and Bell modulations: those are `+MS`'s.)

`ATD`*`string`*

Dials *string* and calls. The dialer sends the digits and `* # A B C D` as DTMF (S11 ms each, as long between them); a comma pauses S8 seconds, `W` two; other characters are skipped. `T` and `P` choose tone or pulse dialing (S16) for the rest of the call. `ATDL` dials the last number again (`Dialing` *number*`...` first). With `X2` or `X4` the modem waits for a dial tone first. The result is CONNECT, BUSY, NO ANSWER, NO DIALTONE or NO CARRIER. In slmodemd the dial string is the SIP call's number (section 9).

ATE[*n*]

Echo of the DTE's characters in command mode: `E0` off, `E1` on (S13; default 1).

ATH[0], ATH1

`H` or `H0` hangs up (on hook); `H1` goes off hook.

ATI[*n*]

Information:

`I0`	the product: `SmartLink Soft Modem`
`I1`, `I2`	the product and version (`SmartLink Soft Modem, 2.9.11`), then `Smart Link Ltd.`
`I3`	the device's name, then the driver's (slmodemd: `/dev/slamr0`, `socket driver`)
`I4`	S0 to S23, eight a line (`s00=000 s01=000 ...`)
`I5`, `I6`	`Stored Profile 0:`, `1:` (there are none)
`I7`	the country: `Country:  USA`

Any other number is ERROR.

ATL[*n*], ATM[*n*]

The speaker's volume, `L0` to `L3` (S30; default 3), and when it is on, `M0` never, `M1` until the call is up, `M2` always (S29; default 1). A SIP line has no speaker: kept for the commands that set them.

ATN[*n*]

Automode (S31; default 1): `N1` lets the modems find their best common modulation from the one `+MS` chose down, `N0` keeps to it. The same as `+MS`'s second parameter.

ATO

Back online after the escape (section 8): CONNECT. ERROR when there is no call to go back to.

ATP, ATT

Pulse or tone dialing (S16; default tone).

ATQ[*n*]

`Q0` gives the result codes, `Q1` keeps them back (S14; default 0).

ATS*n*?, ATS*n*=*v*

Reads S-register n (0 to 255), in decimal on a line of its own, or sets it to v (8 bits: v modulo 256). A number over 255 is ERROR. Section 7 lists them.

ATV[*n*]

`V0` gives the result codes as numbers, `V1` as words (S15; default 1).

ATX[*n*]

Results and dial tone, `X0` to `X4` (S56; default 4):

	CONNECT	before dialing
`X0`	without the rate	no wait for a dial tone
`X1`, `X3`	with the rate	no wait for a dial tone
`X2`, `X4`	with the rate	a dial tone of 1 s, NO DIALTONE if none in S6 seconds (2 s at least)

BUSY (four cadences of 0.2 to 0.8 s) and NO ANSWER come in all of them. A SIP call has no dial tone: slmodemd wants `ATX3`.

ATY[*n*]

`Y0` to `Y3`: kept in S161, nothing more.

ATZ[*n*]

`Z` or `Z0` to `Z3`: hangs up, and the S-registers, `+MS`, modem-on-hold and the data mode (`+FCLASS=0`) as at the start; the country stays. There are no stored profiles: the number changes nothing. OK, and the rest of the line is not read.

5 Ampersand, backslash and percent commands

AT&A[*n*]

&A0 to &A7: the lines before CONNECT (S70; default 0), see the result codes.

AT&C[*n*], AT&E[*n*], AT&H[*n*]

Data carrier detect (&C0, &C1; S60), the rate CONNECT gives (&E0, &E1; S71) and flow control (&H0, &H1; S62): kept, but they change nothing (a pty has no carrier line, CONNECT always gives the receive rate, and flow control is the pty's). Default 0.

AT&D[*n*], AT&F[*n*], AT&R[*n*], AT&S[*n*]

Taken, with or without a digit, and nothing done (ATZ gives the defaults).

AT&K[*n*], AT%C[*n*]

Data compression (S104; default 3): 0 none, 1 to 3 V.44 or V.42bis both ways, as the other modem agrees (V.42 offers both). It needs V.42 (\N).

AT&P[*n*]

&P0 to &P3: the pulse dialing ratio, kept in S28.

AT&V, AT&W

ERROR: there are no stored profiles to show or write.

AT\A[*n*]

\A0 to \A3, MNP's block size: taken (there is no MNP); \A4 and up are ERROR.

AT\N[*n*]

Error correction (S103; default 1): \N0 none; \N1 to \N5 V.42 (LAPM) when the other modem has it, a call without when it does not (V.42's detection phase decides). There is no MNP: an MNP modem that has V.42 connects with LAPM, one with MNP only without error correction or not at all.

AT%E[*n*]

%E0 to %E2, line quality monitoring: kept in S39 (the modulations retrain and renegotiate their rate by themselves).

AT*B

The numbers barred from dialing: none are kept; OK.

AT#CID=*n*, AT#CID?, AT#CID=?

Caller ID between the first two rings: 0 off, 1 or 2 on (=? gives 0,1,2). A SIP call rings without it.

6 Extended commands

6.1 Modulation: +MS

AT+MS=[*mod*][,[*automode*][,[*min*][,[*max*]]]]

The modulation the modem calls or answers with, whether it may fall back (*automode* 1, as N1; S31) and the rates it may connect at, *min* to *max* bit/s (300 to 56000, *min* no more than *max*). A parameter left out keeps its value: AT+MS=,0 only turns automode off. The default is 92,1,300,56000.

mod		
21	ITU V.21	300 bit/s FSK
23	ITU V.23	1200/75 bit/s FSK
103	Bell 103	300 bit/s FSK
22	ITU V.22	1200 bit/s DPSK
212	Bell 212A	1200 bit/s DPSK
122	ITU V.22bis	2400 bit/s, V.22 as its fallback
32	ITU V.32	4800 and 9600 bit/s
132	ITU V.32bis	4800 to 14400 bit/s
34	ITU V.34	2400 to 33600 bit/s, after V.8
90	ITU V.90	as the digital modem, after V.8
91	ITU V.91	two digital modems, after V.8
92	ITU V.92	V.90 with quick connect, PCM upstream and modem-on-hold
56	K56flex	not implemented: runs V.90

V.90 and V.92 are there as the *digital* modem (the server's side, as on a G.711 VoIP line), on an 8000 Hz G.711 line: analogue V.90 and V.92 modems calling connect at up to 56000 bit/s downstream. 90 and 92 offer V.91 too, which two digital modems use instead, and V.34 for the others; on a line at another sample rate they are V.34. With automode V.8 and the V.34 modem hand over to V.32bis (and its V.22bis and V.22) for a modem without V.34, and the answering modem tries V.23 and Bell 103 for one without V.8.

AT+MS?

The settings: `92,1,300,56000`.

AT+MS=?

What may be chosen:
`(21,22,23,122,32,132,34,103,212,56,90,91,92),(0,1),(300-56000),(300-56000)`

6.2 Identification and country

AT+GMI, AT+GMM, AT+GMR

The manufacturer (`Smart Link Ltd.`), the model (`SmartLink Soft Modem`) and the revision (`2.9.11`).

AT+GCI=*h*, AT+GCI?, AT+GCI=?

The country by its T.35 code in hexadecimal (`+GCI:b5`: USA), which sets its dial pause (S8) and hook flash (S24) times; `=?` lists the codes. slmodemd's `-c` option names the country it starts with (`-countrylist` lists them).

6.3 Mode and flow control

AT+FCLASS=*n*, AT+FCLASS?, AT+FCLASS=?

`0` data (default), `1` fax class 1, `8` voice (`=?` gives `0,1,8`).

AT+IFC=2,2

Hardware flow control both ways: taken, as the pty does its own. Any other value is ERROR.

6.4 V.92 modem-on-hold

AT+PMH=*n*, AT+PMH?, AT+PMH=?

`0` lets the other modem put the call on hold, `1` refuses (default 0; `+PMH?` gives `+PMH: 0`).

AT+PMHT=*n*, AT+PMHT?, AT+PMHT=?

How long a hold the modem grants: the code of V.92's Table 33, `0` to `13` (`0` grants none; default 5).

AT+PMHR[=*s*]

In online command mode on a V.92 connection: asks the other modem to go on hold, for *s* seconds (1 to 960; 10 if left out). ERROR otherwise.

6.5 Fax class 1

With `AT+FCLASS=1`, the modem takes T.31's class 1 commands, the fax machine's side being the DTE's (the PBX modules and libdmodem have a T.30 engine of their own as well):

`AT+FTS=`*n*`, AT+FRS=`*n*

Sends silence for, or waits for silence of, *n* times 10 ms.

`AT+FTM=`*m*`, AT+FRM=`*m*`, AT+FTM=?, AT+FRM=?`

Sends or receives data at modulation *m*: CONNECT, then the data with `<DLE>` doubled, ended by `<DLE><ETX>`.

m	
24, 48	V.27ter, 2400 and 4800 bit/s
72, 96	V.29, 7200 and 9600 bit/s
73, 97, 121, 145	V.17, 7200 to 14400 bit/s, long training
74, 98, 122, 146	V.17, 7200 to 14400 bit/s, short training

`AT+FTH=3, AT+FRH=3, AT+FTH=?, AT+FRH=?`

Sends or receives HDLC frames at V.21 channel 2 (300 bit/s), the only one: `=?` gives `3`.

6.6 Voice

With `AT+FCLASS=8` the voice commands of Smart Link's slmodemd are kept (ERROR in the other modes):

command	
`+VIP`	the voice settings as at the start
`+VNH=`*n*	0 or 1
`+VGT=`*n*`, +VGR=`*n*	the gain sending and receiving, 0 to 128
`+VIT=`*n*	the inactivity timer, 0 to 255
`+VSD=`*s*`,`*t*	silence detection: sensitivity and period, 0 to 255
`+VSM=`*m*`,8000`	the compression method, 0 or 1, at 8000 samples/s
`+VLS=`*n*	1 takes the line for voice, 0 leaves it
`+VTX, +VRX, +VTR`	sends, receives, or both
`+VTS=[`*f1*`,`*f2*`,`*d*`]`	a tone of *f1* and *f2* Hz for *d* (0 to 6000)
`+VTS={`*s*`,`*d*`}, +VTS=`*s*	the DTMF digit *s* (for *d*, 0 to 100)
`+VCID=`*n*	as `#CID`

Each takes `?` (its values) and `=?` (their ranges).

7 S-registers

Registers 0 to 255 hold 8 bits each. Those not listed are 0 and free; "kept" means set and read back, with no other effect.

S		default	
0	rings to answer after (0: never)	0	
1	rings so far	0	
2	the escape character	43 (+)	
3	the line's end	13 (CR)	
4	the line feed	10 (LF)	
5	backspace	8 (BS)	
6	wait for a dial tone, s	2	`X2`, `X4`; 2 s at least
7	wait for the answer and the training, s	60	
8	a comma's pause, s	2	the country's

S		default	
9	carrier detect time, 0.1 s	6	kept
10	carrier loss before hanging up, 0.1 s	7	
11	DTMF tone and gap, ms	100	below 40: 95
12	escape guard time, 20 ms	50	kept: the guard is 0.5 s
13	echo (`E`)	1	
14	quiet (`Q`)	0	
15	verbose (`V`)	1	
16	tone (1) or pulse (0) dialing (`T`, `P`)	1	
24	hook flash time, 10 ms	20	the country's
28	pulse ratio (`&P`)	0	kept
29	speaker control (`M`)	1	
30	speaker volume (`L`)	3	
31	automode (`N`, `+MS`)	1	
32	the modulation (`+MS`)	92	
39	line quality (`%E`)	0	kept
56	results and dial tone (`X`)	4	
60	carrier detect (`&C`)	0	kept
62	flow control (`&H`)	0	kept
70	CONNECT's lines (`&A`)	0	
71	CONNECT's rate (`&E`)	0	kept
92	the answer's delay, s (0: 10 ms)	2	before the answer tone
103	error correction (`\N`)	1	
104	data compression (`%C`, `&K`)	3	
161	`Y`	0	kept

8 Online, the escape and hanging up

Online, what the DTE sends goes to the other modem, except the escape: three S2 characters (`+++`), each within 0.5 s of the one before, with 0.5 s of nothing before them and after them. The modem then says OK and is in online command mode, the call still up: `ATO` goes back online, `ATH` hangs up, and on a V.92 connection `AT+PMHR` asks for modem-on-hold.

The call ends with NO CARRIER when the other side hangs up or its carrier is gone for S10 tenths of a second. slmodemd hangs up as well when the DTE drops DTR (it sets the pty's speed to 0) or, with HUPCL set on the pty, closes it; it then makes a new pty (`/dev/ttySL0` follows it).

9 slmodemd on a SIP line

- `ATD`*number* calls `sip:`*number*`@`*domain* of `SIP_LOGIN`. A telephone number (only V.250's dial characters: digits, `* # + A B C D`, the modifiers `T P W , @ !` `;` and `- ( ) .` and blanks) goes as its digits and DTMF characters: `ATDT555-1000` calls `sip:5551000@`*domain*. Anything else is a SIP user's name, as it is: `ATDalice`.
- The line has no dial tone: `ATX3` (or `X1`), else NO DIALTONE.
- With `-L` slmodemd registers and takes calls: RING every 3 seconds while one waits, `ATA` or S0 rings (`ATS0=1`) answer it, and the rings stop if the caller gives up. A call that comes while the line is in use is refused (486, busy); without `-L` every call is (480).
- The call is G.711 at 8000 Hz, which is what V.90 and V.92 need (`AT+MS=92`, the default). Over a real PSTN gateway a lower modulation (`AT+MS=132,0,4800,9600`) may do better.

10 Differences from Smart Link's ST56 command set

- No stored profiles: `&W` and `&V` are ERROR, `&F` and `Z`*n*'s number do nothing; `ATZ` gives the defaults.
- No MNP (classes 2 to 5, 10, 10EC): `\A` is taken and `\N1` to `\N5` are all V.42; Rockwell's MNP 10 and 10EC commands (`-K`, `-SEC`, `)M`, `@M`, `*H`, `-Q`) are ERROR. A modem with MNP 10EC connects with V.42 (its "EM" detection patterns are passed over). See `docs/link.md`, MNP.

- K56flex is not there: `+MS=56` runs V.90.
- V.90 and V.92 as the digital modem, V.91, V.44 and V.92's modem-on-hold (`+PMH`, `+PMHT`, `+PMHR`) are D-Modem's.
- `*B`, `\A` and `+M`*x* give their own result (OK, or ERROR for `+M`*x* other than `+MS`), not the command before them.
- Kept settings that change nothing on a pty or a SIP line: `&C`, `&E`, `&H`, `%E`, `&P`, `L`, `M`, `Y`; S9 and S12.